NAME: ______________________ **PERIOD:** ________ **DATE:** ____________

Which Way Does the Condition Point?

AP Statistics · Unit 2 · Topic 2.6 · B: 2026-10-29 · E: 2026-10-30

[STAR] TODAY'S PROBLEM

Suppose a school screens 1,000 students for a hypothetical condition. Exactly 5% have the condition. The test has 90% sensitivity: among students with the condition, 90% test positive. Its false-positive rate is 10%: among students without the condition, 10% test positive. The implied counts are shown.

Mina: "The test catches 90% of students who have the condition, so a positive result means a student has a 90% chance of having it."

Luis: "We need to count everyone who tested positive before finding the chance of having the condition."

Screening results (counts)

Status	Pos.	Neg.	Total
Condition	45	5	50
No condition	95	855	950
Total	140	860	1,000

FIRST TAKE — before we discuss (5 min)

Does a positive result mean a 90% chance of having the condition? Choose a claim and cite one table count.

[BOOK] THE GIVEN EVENT SETS THE DENOMINATOR (keep on the page)

$P(A \mid B)$ means the probability of A **given** B. Restrict attention to the outcomes in B: $P(A \mid B) = \dfrac{P(A \cap B)}{P(B)} = \dfrac{\text{count in both } A \text{ and } B}{\text{count in } B}$. In general, $P(A \mid B) \neq P(B \mid A)$ because the denominators are different. **Sensitivity** is $P(\text{positive} \mid \text{condition})$. The **false-positive rate** is $P(\text{positive} \mid \text{no condition})$.

Work (a)–(c) from the rules box:

DOK 1 **(a)** For $P(\text{condition} \mid \text{positive})$, identify the numerator and denominator counts.

DOK 2 **(b)** Find $P(\text{positive} \mid \text{condition})$ and $P(\text{condition} \mid \text{positive})$. Show each count fraction and percent.

DOK 3 **(c)** Is Mina's claim supported? Use the positive-test counts for students with and without the condition. Explain how reversing given changes the group in the denominator. Write 3–4 sentences.

Frame: The 90% uses the group ____________________, but a positive result directs us to _______________________.

Frame: Among _______ positive results, _______ have the condition, so the chance is _________.

EXIT — turn this in

One thing the rules box changed about my first take: ____